
Supplementary information

Imaging the Meissner effect in hydride superconductors using quantum sensors

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Supplementary information: Imaging the Meissner effect in hydride superconductors using quantum sensors

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(Dated: December 5, 2023)

THE NV CENTER

The NV center in diamond is a crystallographic defect comprising a substitutional nitrogen atom adjacent to a lattice vacancy [1]. The electronic ground state of the negatively charged NV center is spin triplet ($S = 1$). In the absence of external perturbations, the ground-state spin Hamiltonian is given by $H = D_{\text{gs}} S_z^2$, where $D_{\text{gs}} = (2\pi) \times 2.87$ GHz is a temperature-dependent zero-field splitting between the $|m_s = 0\rangle$ spin sub-level and the degenerate $|m_s = \pm 1\rangle$ spin sub-levels, and $\{S_x, S_y, S_z\}$ are spin-1 operators quantized along the N-V axis ($\hat{z}$). The quantization axis may be oriented along any of the diamond bonds resulting in four subgroups of NV centers. A magnetic field $\vec{B}$ couples with the Hamiltonian $H_B = \gamma_B \vec{B} \cdot \vec{S}$, where $\gamma_B = (2\pi) \times 2.8$ MHz/G is the NV's gyromagnetic ratio and $\vec{B}$ is expressed in the local frame of the NV center [1, 2].

A stress tensor, σ , couples with the Hamiltonian [3, 4],

$$H_s = \Pi_z S_z^2 + \Pi_x (S_y^2 - S_x^2) + \Pi_y (S_x S_y + S_y S_x) \quad (\text{S1})$$

where $\Pi_i = \Pi_i(\sigma)$ are functions of the stress tensor as follows:

$$\Pi_z = \alpha_1 (\sigma_{xx} + \sigma_{yy}) + \beta_1 \sigma_{zz} \quad (\text{S2})$$

$$\Pi_x = \alpha_2 (\sigma_{yy} - \sigma_{xx}) + \beta_2 (2\sigma_{xz}) \quad (\text{S3})$$

$$\Pi_y = \alpha_2 (2\sigma_{xy}) + \beta_2 (2\sigma_{yz}) \quad (\text{S4})$$

are expressed in terms of the stress components in the NV's local frame. Here, $\{\alpha_1, \beta_1, \alpha_2, \beta_2\} = (2\pi) \times \{8.6(2), -2.5(4), -1.95(9), -4.50(8)\}$ MHz/GPa are the stress susceptibilities. Stress components that preserve the C_{3v} -symmetry of the NV center maintain the degeneracy of the $|m_s = \pm 1\rangle$ spin-sublevels and induce an overall energy shift of Π_z . Stress components that break C_{3v} -symmetry lift the degeneracy between the $|m_s = \pm 1\rangle$ spin-sublevels and induce a splitting $2\Pi_{\perp} = 2\sqrt{\Pi_x^2 + \Pi_y^2}$. In addition, symmetry breaking stress components mix the $|m_s = \pm 1\rangle$ states into eigenstates $|\pm\rangle = (|m_s = 1\rangle \pm e^{i\phi_{\Pi}} |m_s = -1\rangle)/\sqrt{2}$, where $\phi_{\Pi} = \arctan(\Pi_y/\Pi_x)$.

We excite electronic transitions in the NV center using laser light with wavelength $\lambda = 532$ nm. Radiative transitions involved in electronic excitation and fluorescent relaxation in the NV center are spin conserving processes [1, 5]. In addition to these radiative transitions, the NV center also shows a non-radiative inter system crossing (ISC) mechanism, whereby the $|m_s = \pm 1\rangle$ spin sub-levels of the electronic excited state can transition into the singlet manifold. Further non-radiative relaxation from the singlet manifold to the electronic ground state is not highly spin selective [1, 5]. This leads to two important consequences. First, it enables optical polarization to the $|m_s = 0\rangle$ spin sub-level under continuous laser excitation. Second, it causes a reduction in the fluorescence rates of the $|m_s = \pm 1\rangle$ spin sub-levels with respect to the $|m_s = 0\rangle$ spin sub-level. Indeed, signal contrast of an ODMR measurement depends on the difference between the fluorescence intensity of these spin sub-levels. As we will discuss in the next section, the ISC mechanism — and thus the ODMR contrast — is sensitive to the applied stress and can be significantly altered at high pressures.

In addition to ODMR measurements, where one measures DC magnetic fields through the Zeeman splitting of the ODMR resonances, the NV center can also be used to measure AC magnetic fields. Here, protocols inspired by pulsed nuclear magnetic resonance (NMR) techniques can be translated to the case of electron spin resonance of the NV center [6]. The main idea of these AC magnetometry techniques is to first, create a coherent superposition of spin states and subsequently, use pulse sequences to lock the evolution of the NV's electron spin to a specific frequency ω . A fluorescence contrast measurement is used to read out the final state of the NV, allowing for the determination for AC magnetic fields of the form $Be^{i\omega t}$. In this work, we demonstrate a spin echo measurement at megabar pressures [maintext Fig. 1(e)].

NV sensing at megabar pressures

The extension of NV sensing to megabar pressures has been a long-standing challenge due to several factors: reduction in NV fluorescence, inhomogenous broadening of ODMR spectra, and significant loss of ODMR contrast [7, 8]. In this section, we compare NV measurements under pressure for [100]-cut and [111]-cut anvils [Fig. S1].

For studies in a [100]-cut anvil, we use a type Ib diamond with 200 μm flat culet having a shallow layer of NV centers incorporated up to 50 nm below the surface (recipe included in the following section) [3]. We use an untreated type Ia diamond as the opposing anvil, a potassium chloride (KCl) pressure medium, and a combination of ruby and culet raman spectra for pressure calibration [9, 10]. We refer to the lab frame by $\{X, Y, Z\}$ where Z is the loading axis, and to the local frame of the NV subgroups by $\{x, y, z\}$.

Using the NV center, we measure the dominant culet stresses, σ_{ZZ} and $\sigma_{\perp}$. The former, σ_{ZZ} , is the loading stress, and the latter, $\sigma_{\perp} = (\sigma_{XX} + \sigma_{YY})/2$, is a biaxial in-plane stress, which at high pressures is related to cupping of the diamond culet [11]. For a [100]-cut anvil, these stresses project degenerately across all four NV subgroups, having both symmetry preserving and symmetry breaking components (Fig. 1(b) in maintext). In particular, the measured stress parameters $\{\Pi_z, \Pi_{\perp}\}$ map to $\{\sigma_{ZZ}, \sigma_{\perp}\}$ stresses as follows:

$$\Pi_z = \frac{1}{3}(2\alpha_1 + \beta_1)(2\sigma_{\perp} + \sigma_{ZZ}) \quad (\text{S5})$$

$$\Pi_{\perp} = \frac{2}{3}|(\alpha_2 - \sqrt{2}\beta_2)(\sigma_{ZZ} - \sigma_{\perp})| \quad (\text{S6})$$

We use this relationship to extract the stresses from the NV's ODMR spectrum measured at the center of the culet [Fig. S1(c)]. The loading stress, σ_{ZZ} , agrees closely with the sample pressure which is calibrated independently (using ruby and culet raman pressure markers) [9, 10].

ODMR contrast—For the [100]-cut anvil, we observe a significant loss in the ODMR contrast with increasing pressure [Fig. S1(a)]. In addition, at pressures above ~ 50 GPa, we observe a surprising inversion of ODMR contrast for one of the resonances [Fig. S1(b)]. However, for the [111]-cut anvil, neither of these effects are observed. Notably, for this anvil cut, the ODMR contrast at pressures up to ~ 140 GPa remains comparable to that typically achieved under ambient conditions [Fig. S1(d)].

A microscopic understanding of the observed changes in the ODMR contrast requires a careful analysis of the NV center's ISC mechanism in the presence of stress [5]. While this is the subject of ongoing studies [12], we summarize our preliminary findings as follows. C_{3v} -symmetry preserving stresses induce an overall quantitative change in the ISC rate. Interestingly, a mixture of hydrostatic stress and an uniaxial stress along the quantization axis ($\hat{z}$) of the NV center leads to the highest ISC rate at megabar pressures. This condition is approximated in a [111]-cut anvil

for one of the NV subgroups and may explain the retention of ODMR contrast at these pressures [Fig. S1(d)] [3]. In comparison, C_{3v} -symmetry breaking stresses can allow new non-radiative ISC transitions from the spin manifold in the electronic excited state, thereby inducing qualitatively different changes to the ISC mechanism. In a [100]-cut anvil, the presence of symmetry breaking stresses may be responsible for both contrast loss as well as the appearance of positive contrast peaks.

EXPERIMENTAL DETAILS

Sample preparation

For NV measurements, we use 16-sided standard design diamond anvils with [111]-crystal cut culets (Syntek Co. Ltd. and Almax-easyLab). These anvils are polished from synthetic type-Ib ($[N] \leq 200$ ppm) single crystal diamonds. The culet diameter is 100 μm with a bevel at 8.5° up to 300 μm . Similar to previous work, we perform $^{12}\text{C}^+$ ion implantation at an energy of 30 keV with a dosage of $5 \times 10^{12} \text{ cm}^{-2}$ (CuttingEdge Ions, LLC) to generate a layer of vacancies up to 50 nm from the culet surface [3]. Following implantation, we vacuum anneal the diamond anvils (at $P < 10^{-6}$ mbar) in a home built furnace at a temperature $> 850^\circ\text{C}$ for 12 hours. During the annealing step, the vacancies become mobile in the lattice and can probabilistically form NV centers by migrating to lattice sites adjacent to existing substitutional nitrogen defects. The NV diamond is glued to the cylinder side of a 23-mm miniature panoramic diamond anvil cell (DAC) machined out of non-magnetic NiCrAl alloy and beryllium copper (Cu-Be) [13]. A platinum (Pt) wire with a diameter of 20 μm is incorporated near the culet to apply microwave (MW) radiation for spin state control. During compression, the Pt wire on the diamond bevel gradually expands into a foil. On the opposing (piston) side, we glue a type-Ia diamond anvil with [100]-crystal cut. On this anvil, we sputter Molybdenum (Mo) leads for performing electrical resistance measurements. A double-sided insulating gasket, fashioned out of rhenium (Re) foil and calcium fluoride (CaF_2) mixed with epoxy, is used to isolate the transport leads and the MW wire.

We synthesize cerium hydride following the recipe detailed in [14]. We load cerium (Ce) metal (99.8% purity) and ammonia borane (NH_3BH_3) in an argon (Ar) glove box. We first compress the loading to pressures above 100 GPa. The pressure P is calibrated using the raman line of the [111]-cut culet [see Fig. S2(a,b)] [15]. Following this, we perform single-sided laser heating by focusing a 1070 nm laser beam to a $\sim 3 \mu\text{m}$ spot at different locations on the Ce sample. Laser heating induces the decomposition of ammonia borane into cubic boron nitride (cBN) and hydrogen. The released hydrogen subsequently reacts with the cerium to form cerium polyhydrides. ($\text{Ce} + \text{NH}_3\text{BH}_3 \rightarrow \text{CeH}_x + \text{cBN}$).

After synthesis, four point resistance is measured using the delta technique [16] in a liquid He (wet) cryostat (detailed in [14]). We use a Keithley 6221/2182A setup for delta measurements. The 6221 provides a square-wave current output where the direction of the current alternates within each cycle. The 2182A performs voltage measurements synchronously with the current source. A three-point moving-average algorithm is used to calculate the voltage readings to remove the effects drift in thermal EMFs in the measurement leads. T_c is determined by the onset of the resistance drop. We measure $T_c \approx 92$ K for S1 at $P \approx 118$ GPa and $T_c \approx 91$ K for S2 at $P \approx 137$ GPa indicating the formation of CeH_9 in both samples [see Fig. S2(c,d)]. T_c is suppressed upon the application of tesla scale magnetic fields [insets in Fig. S2(c,d)]. For both samples S1 and S2, we find that the change in T_c is less than 0.5% for an applied field $H < 1000$ G. This allows us to calibrate for temperature gradients in the dry cryostat where NV measurements are performed up to $H_z = 240$ G fields [Fig. S5].

Note: Sample S1 failed before all studies could be completed and sample S2 failed shortly after the first submission of our manuscript.

Note: For electrical resistance measurements using the delta technique we vary sample temperature at the rate of 1 – 2 K/min to allow for sample thermalization. For sample S2 at zero field, we measure a temperature hysteresis of 0.5 K [Fig. S2(h)] between the cooling and warming cycles which is consistent with previous studies of CeH_9 [14].

Note: Since the synthesized superconducting CeH_9 accounts for a small fraction of the inhomogeneous high pressure sample. However, we can use the Van der Pauw technique to roughly estimate the resistivity of the sample in the superconducting state (below T_c). Assuming a sample thickness of $d \sim 1 \mu\text{m}$ and measuring the resistance R at the lowest temperatures, the resistivity is then approximated by $\rho = \pi d R / \ln 2$ [17]. Using this method, we measure $\rho \approx 1.1 \times 10^{-9} \Omega\text{m}$ for sample S1 and $\rho \approx 1.4 \times 10^{-11} \Omega\text{m}$ for sample S2. The measured resistivity is a factor of 10 and a factor of 10^3 smaller than that of copper for samples S1 and S2 respectively [see Fig. S2(g)].

Apparatus for simultaneous ODMR and electrical resistance measurements

We perform NV measurements in a dry closed cycle cryostat (Attocube AttoDRY800) integrated with home-built confocal fluorescence microscope (similar to previous work [3]). A galvanometer (Thorlabs GVS212) is used to raster the excitation beam from a 532 nm laser (Sprout Solo-10W). The excitation light is switched by an acousto-optic modulator (AOM, Gooch & Housego AOMO 3110-120) in double pass configuration. A 4*f*-telescope focuses the laser beam at the entrance of an objective lens (Mitutoyo LWD NA 0.7) that is used to image the diamond culet. The fluorescence collected by the objective is separated from the excitation path by a dichroic mirror and measured by a fiber-coupled single photon counting module (Excelitas SPCM-AQRH-64FC). A data acquisition device (National Instruments USB-6343) is used for fluorescence counting as well as modulation of a microwave source (SRS SG386) for continuous wave ODMR measurements. For pulsed measurements, a programmable multi-channel pulse generator (SpinCore PulseBlasterESR-PRO 500) is used to gate the AOM switch and the microwave source (SRS SG386). For both continuous wave and pulsed measurements, the microwaves are amplified using a Minicircuits ZHL-25W-63+ amplifier and subsequently channeled to the sample using coaxial cables. The Pt-wire at the sample is connected to the coaxial terminals using 100 μm copper magnet wires. An external magnetic field is applied by running a current I from a DC power supply (BK Precision 9103) into a custom built electromagnet (Woodruff Scientific) that sits outside the cryostat shroud [Fig. S4(a)]. A lock-in amplifier (Zurich Instruments MLFI DC-500kHz) in combination with a voltage-controlled current source (Stanford Research Systems CS580) is used to measure four point resistance of the sample. In this setup, we are limited to applied fields of ≈ 240 G. Electrical resistance measurements at high fields [Fig. S2(c,d)] are performed in a separate cryostat [14].

In sample S2, we find that CeH₉ region is synthesized directly on top of two transport leads [Fig. S3(b)]. For four point resistance measurements, we pick up voltage between these two transport leads to get near-zero resistance for $T < T_c$.

Creation of NV centers during laser heating

In addition to the layer of NV ensembles created by ion implantation and annealing, we find that a dense ensemble of NV centers are created by laser annealing of the diamond during CeH₉ synthesis. In particular, regions of the sample that are laser heated show noticeably higher fluorescence in both samples S1 and S2. Fig. S3 (a,d) illustrates this for sample S2: we see micron-scale bright spots resembling foci of an optical beam. The axial (Z) resolution of $\sim 10 \mu\text{m}$ in our confocal microscope is considerably poorer than the sub-micron transverse (XY) resolution; therefore, it is difficult to directly verify the depth and distribution of the NV centers in the laser heated regions. However, a comparison of the XZ scans shows that the depth and distribution of the implanted NV layer and the laser-heating-created NV layer are comparable [Fig. S3(g,h,i)] suggesting that the laser-heating-created NV layer is likely $\lesssim 1 \mu\text{m}$ deep. This is also consistent with an analysis of the broadening of the ODMR spectrum [Fig S3(e,f)]; indeed, if the laser-heating-created NV layer was thicker, one would expect a significantly broader ODMR spectrum in this region owing to the variation of the magnetic field over the thickness of the layer.

We note the following remarks. First, due to compression of the sample to megabar pressures, we expect significant contact stress at the interface between the diamond and the Ce sample. Indeed, we find that ODMR spectra in regions without laser heating exhibit large splittings ($2\Pi_{\perp} \gtrsim (2\pi) \times 200$ MHz) [blue data in Fig. S3(c)]. However, ODMR spectra at laser heated regions show appreciably smaller splittings ($2\Pi_{\perp} \sim (2\pi) \times 100$ MHz) [green data in Fig. S3(c)]. This may be due to local stress relaxation because of the high temperatures generated during laser heating [18]. Second, laser heated regions of the culet show markedly higher ODMR contrast compared to other regions [Fig. S3(c)].

Calibration of external magnetic field H_z

At the beginning of every cooldown, we calibrate the external magnetic field (H_z) as a function of the current applied to the electromagnet (I). We achieve this by measuring the NV's ODMR splitting Δ as a function of I . Our calibration necessarily assumes that, at temperatures $T > T_c$, the sample does not contribute to the local magnetic field, B_z , and thus $B_z = H_z$ (in Gaussian units). Knowing $B_z = H_z \propto I$ and $\Delta = \sqrt{(2\Pi_{\perp})^2 + (2\gamma_B B_z)^2}$, each calibration yields the proportionality constant dH_z/dI . Due to the large size of the electromagnet coil (~ 0.5 m)

compared to the diamond culet (100 μm), we expect the variation of applied field over our sample region to be small. [Fig. S4(a)]. We verify this by performing the calibration at multiple spatial locations and also at several temperatures for the same spatial location over the course of several cooldowns [Fig. S4(b-c)]. Minor variations in the calibration (dH_z/dI) between cooldowns result from differences in the placement of the coil around the cryostat's vacuum shield during experimental setup. Nevertheless, for each calibration we find that spatial variations in H_z across the sample region is around $\sim 1\%$ [see data for cooldown 6 in Fig. S4(c)].

Correction for thermal gradients

Following CeH_9 synthesis, four point resistance measured via delta technique in a Liq. He (wet) cryostat reflects the true T_c for both samples [Fig. S2(c,d)]. NV measurements are performed in a dry closed cycle cryostat where the temperature is measured at the cold finger and not directly on the DAC body. In dry cryostats, sample thermalization can pose a significant challenge due to limited thermal contact and large thermal mass of the DAC. We incorporate two additional thermal links: the first, between the cell body and cold finger (custom fabricated by FourNineDesign) and, the second, between the gasket and the cell body (cut out of indium foil). Despite these attempts, we measure a temperature gradient, ΔT , between the temperature recorded at the cold finger and the true sample temperature. For both samples S1 and S2, four point resistance measured via lock-in technique in the dry cryostat shows transitions at temperatures lower than the true T_c [Fig. S5(a,b)].

We determine ΔT by measuring the shift of the resistance drop between the four point lock-in and delta measurements. This corresponds to a peak in dR/dT as shown in the insets of Fig. S5(a,b). We estimate $\Delta T \approx 14.2$ K for S1 and ΔT between ≈ 15.7 K and ≈ 21.2 K for S2 across different cooldowns. For clarity, we present sample temperature T in our data by simply adding this correction to the temperature read out at the cold finger.

We note the following remarks concerning the accuracy of this correction. First, minor adjustments of the sample and the thermal link between cooldowns can change this temperature gradient (as seen in the case of S2). In the case of S1, due to coupling between the transport leads and the Pt wire, four point resistance could only be measured when the sample was disconnected from the microwave circuitry used for ODMR measurements. This isolation of the system may lower the temperature gradient, leading to different temperature corrections (ΔT) for ODMR and electrical resistance measurements. In Fig. E1(e) (methods), the apparent separation between the magnetic and resistive transitions is likely due to this reason. Second, the temperature gradient may itself be temperature dependent and our correction does not account for this. Third, when sweeping temperatures across T_c , the rate of sample thermalization may depend on the direction of the sweep (i.e., field heating as opposed to field cooling) [see Fig. S5(c,d)]. Fourth, during NV measurements in the dry cryostat, four point resistance is often measured under an applied field $H_z < 300$ G. We neglect the modest suppression (less than 0.1%) of the true T_c of the samples when estimating ΔT [see insets of Fig. S2(c,d)]. Finally, we measure a temperature hysteresis in sample S2 where T_c differs by less than 0.5 K between cooling and warming cycles [Fig. S2(h)]. This naturally translates to an error of ~ 0.5 K in our correction.

This method for correction of thermal gradients assumes that T_c itself does not change over the course of the experiment; below, we provide some anecdotal evidence for the same. First, we look at Fig. S5(c-d) which show electrical resistance measurements on field heating (after zero field cooling) and field cooling respectively. This data set, taken continuously over the course of a month in the dry cryostat, shows a difference in T_c of at most ~ 4 K after many thermal cycles; it is safe to say that T_c is stable to within ~ 4 K over this time frame. Although, the measurements were taken for fields ranging from $H_z = 34$ G to $H_z = 206$ G, as discussed earlier the suppression of T_c at these small fields should be negligible. Rather, we attribute this ~ 4 K deviation to incomplete thermalization of the sample during the different field heating sequences. We compare this to a less than 0.5 K difference in T_c measured in the wet cryostat on thermal cycling at zero field [Fig. S2(h)]. The latter is consistent with previous measurements on CeH_9 which suggests that changes in T_c should be restricted to a few kelvin over the course of a few months of data taking. Second, changes in T_c may be induced by structural transitions or significant changes in the sample pressure. In the former case, volume changes accompanying structural transitions would change the sample pressure and thereby shift the ODMR resonances at the culet. In the latter, case one would observe a similar effect on the ODMR resonances. We can rule out these events in our experiment because we did not measure any such changes in the NVs ODMR resonances over the course of our experiment.

The effect of temperature

The temperature dependence of the zero-field splitting leads of a difference of $\approx (2\pi) \times 7.3$ MHz in the value of D_{gs} between room temperature ($T = 300$ K) and cryogenic temperatures ($T = 4$ K) [19]. In comparison, we observe significant stress induced shifts ($\Pi_z > (2\pi) \times 1$ GHz) and splittings ($2\Pi_\perp \sim (2\pi) \times 100$ MHz) in the ODMR spectrum at megabar pressures. Furthermore, given that we only interpret the ODMR splitting caused by stress and magnetic fields, we can exclude the direct effect of temperature in our measurements. However, this does not preclude indirect effects, including a temperature dependent change in the stress due to thermal expansion or contraction in the DAC body (discussed below).

A comment on errors

In our calculation of B_z , we perform error propagation of fitting error for gaussian fits to the ODMR spectra. These errorbars are included in all our plots. We note that this may not necessarily reflect the true error in our measurements. In this section, we list the different sources of error.

- A change in temperature induces volumetric changes in the DAC body [13]; this results in different stress parameters $\{\Pi_z, \Pi_\perp\}$ for different temperatures [Fig. S6]. The extraction of B_z at a specific temperature relies on the accurate determination of the stress splitting ($2\Pi_\perp$) at the same temperature. Most of these studies are performed on sample S2, where we find that temperature dependent changes in the $\Pi_\perp$ splitting are within $\approx (2\pi) \times 8$ MHz [Fig. S6(b)]. However, it was not feasible to determine the temperature dependence of $2\Pi_\perp$ at all relevant spatial locations. For each spatial location in Fig. 4(e-h), we extract B_z at all temperatures points by using the $2\Pi_\perp$ splitting measured at a single temperature point ($T = 86$ K) at the respective spatial location. The systematic disagreement between B_z and H_z for $T > T_c$ for this data set is likely due to minor changes in the $2\Pi_\perp$ parameter across the temperature sweep. In particular, an error of $(2\pi) \times 8$ MHz in the $\Pi_\perp$ parameter translates to an error of 1.7 G in the determination of B_z for this study. Owing to the same reason, we find similarly small systematic disagreements for data shown in Fig. E9 (methods). This issue may be mitigated by switching to widefield microscopy [3, 20, 21], where one can determine the temperature dependence of $\{\Pi_z, \Pi_\perp\}$ across the entire culet more efficiently.
- We note that laser heating at high pressure (> 100 GPa) results in spatial variation of $2\Pi_\perp$ values laterally across the sample region. Despite careful monitoring, small variations in the measurements may be induced by spatial (XY) drifts in the sample during data acquisition. In almost all cases, the extracted signal magnitudes outweigh errors introduced in this process. Image registration in widefield datasets can correct for these errors in future [22].
- ODMR spectra in the laser heated regions sample the stress gradient over the depth of order $\sim 1 \mu\text{m}$ [Fig. S3(d,i)]. Owing to this, we do not observe clear gaussian lineshapes at most spatial locations in these regions. To simplify our analysis, we restrict our studies to spatial locations where the ODMR lineshapes are approximately gaussian.

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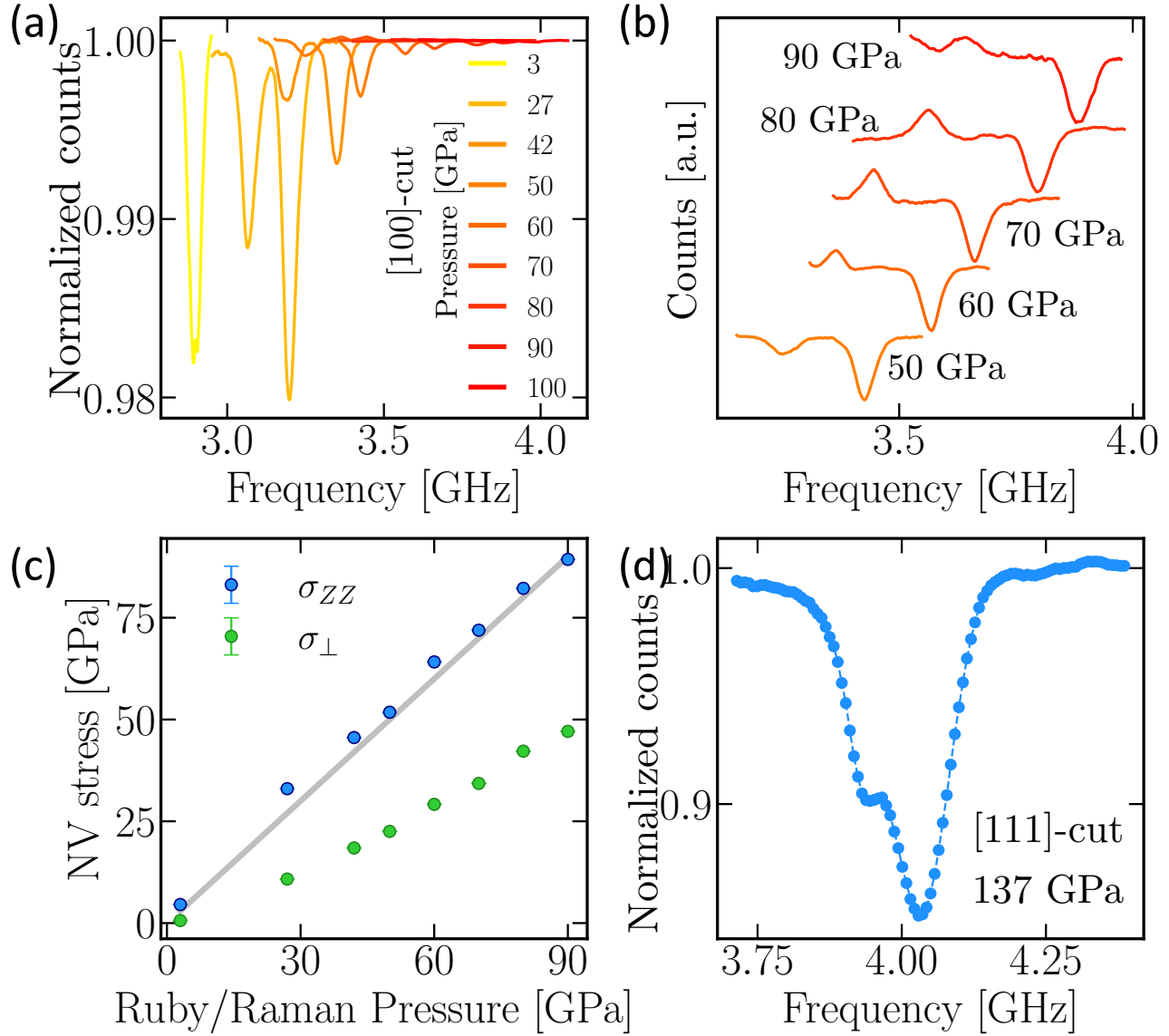


FIG. S1. **Pushing NV sensing to megabar pressures.** (a) ODMR measurements under pressure in a [100]-cut anvil exhibit a drastic reduction in contrast with increase in pressure. The dominant culet stresses $\{\sigma_{ZZ}, \sigma_{\perp}\}$ have degenerate symmetry-preserving and symmetry-breaking projections on each NV subgroup, thereby inducing both a shift, Π_z , and a splitting, $2\Pi_{\perp}$, with increasing pressure. (b) In a [100]-cut anvil, we see a surprising inversion of contrast for one of the ODMR peaks at pressures above ~ 50 GPa. (c) In a [100]-cut anvil, we see good agreement between the values of the loading stress σ_{ZZ} extracted from NV measurements (blue points) and values of the sample pressure calibrated via a combination of ruby and culet raman spectra [9, 10]. For comparison, $x = y$ line is plotted in grey. We also measure an increase in the $\sigma_{\perp}$ stress (green points) consistent to cupping of the diamond culet [11]. (d) In a [111]-cut culet (sample S2), we show $\sim 15\%$ contrast at 137 GPa pressure for the [111] NV subgroup.

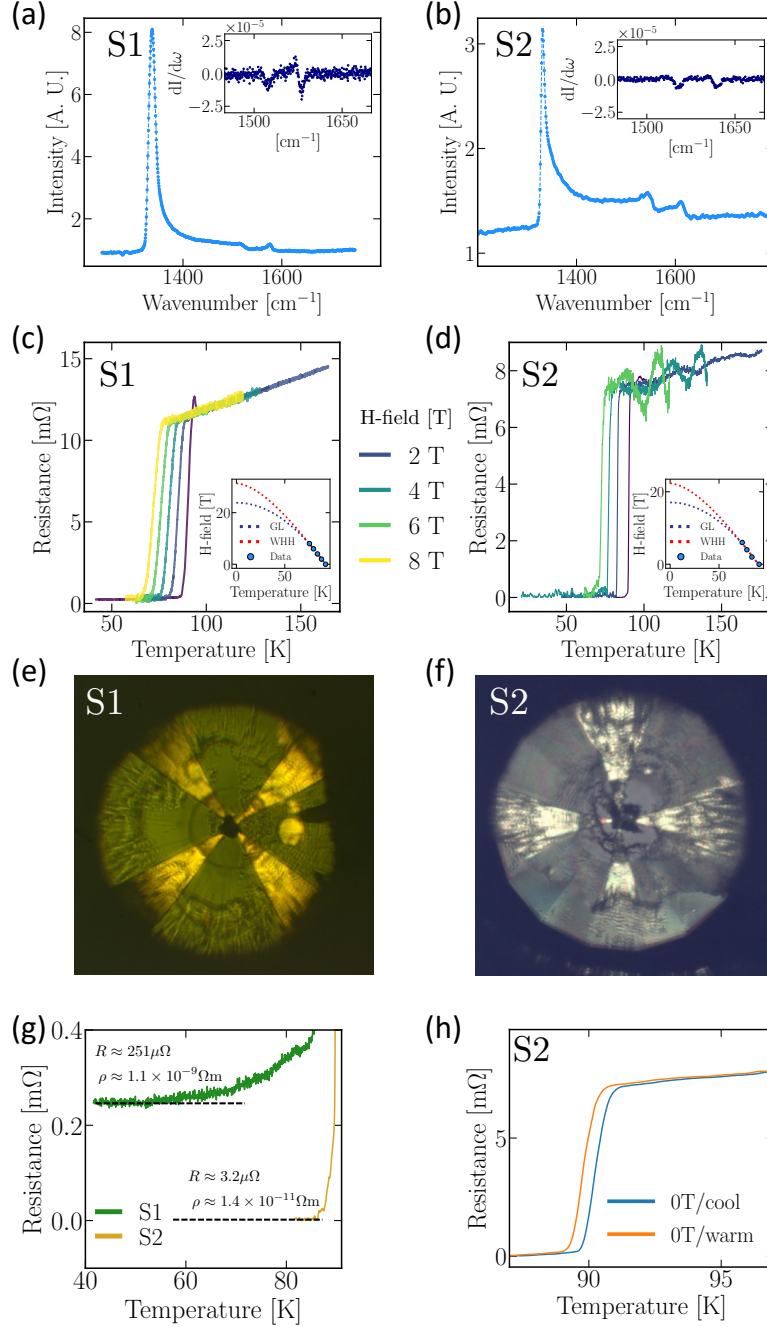


FIG. S2. Pressure calibration and verification of CeH_9 synthesis. Following the procedure detailed in [15], we use culet raman spectra for samples (a) S1 and (b) S2 to determine loading pressure. The insets plot the derivative of the raman fluorescence intensity; we ascertain pressures of ~ 118 GPa for sample S1 and ~ 137 GPa for sample S2 respectively. Following synthesis through laser heating, four point resistance of (c) S1 and (d) S2 were measured via delta technique [16]. The measured $T_c = 94$ K for S1 and $T_c = 91$ K for S2 suggest the formation of CeH_9 in both samples [14]. T_c is determined by onset of the resistance drop. Insets depict the dependence of T_c on the applied field H (blue data points). Within the parameter regime of operation, the measurements show good fits to both Ginzburg-Landau (GL) theory (blue dotted line) [23] and Werthamer-Helfand-Hohenberg (WHH) theory (red dotted line) [24, 25]. (e-f) White light microscope images of samples S1 and S2 showing the high pressure sample and transport leads. The Pt wire for MW delivery casts a shadow on the right side on (e) and can be seen at bottom of (f). (g) Using Van der pauw technique, we can estimate sample resistivity from the resistance measured below the transition temperature T_c . Assuming a sample thickness of $1 \mu\text{m}$, we can get $\rho \approx 1.1 \times 10^{-9} \Omega\text{m}$ for sample S1 and $\rho \approx 1.4 \times 10^{-11} \Omega\text{m}$ for sample S2 [17]. (h) For sample S2, we measure a temperature hysteresis in T_c that is less than 0.5 K at zero field in the wet cryostat.

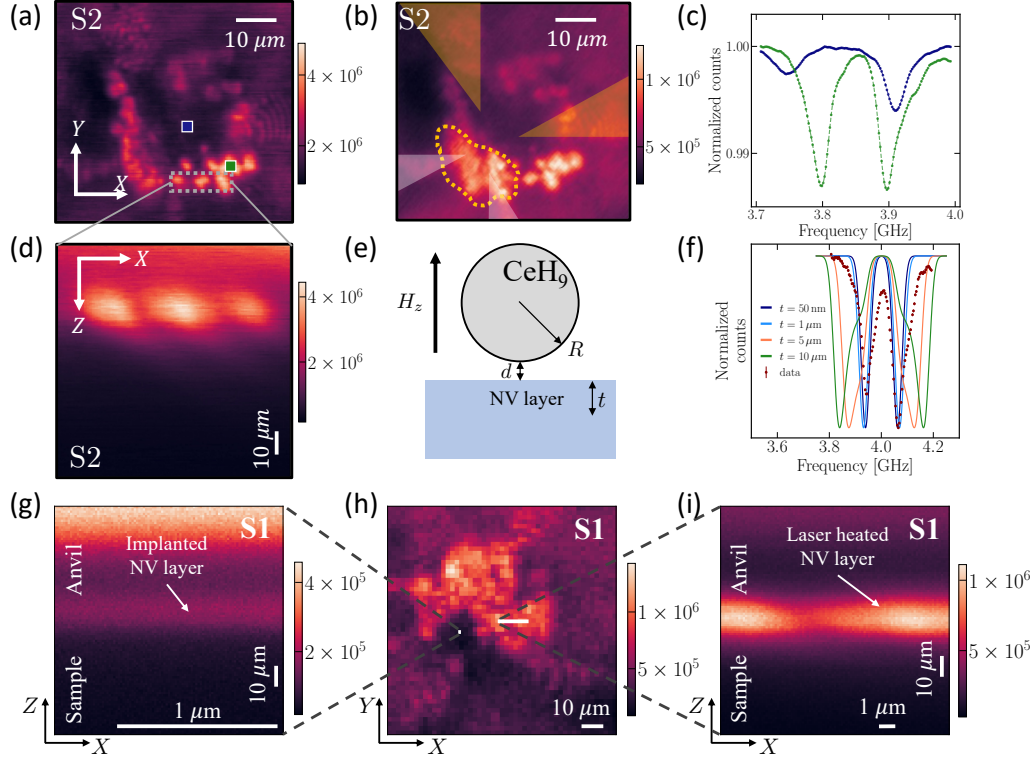


FIG. S3. Creation of NV centers at the culet during laser heating. (a) A confocal fluorescence image of sample S2 after an initial round of laser heating shows bright regions with higher NV fluorescence suggesting the creation of NV centers during laser heating for superhydride synthesis [26]. (b) A confocal fluorescence image of sample S2 after a second round of laser heating within the region enclosed by a dotted yellow line clearly shows the creation of more NV centers in this region [compared to fluorescence scan in (a)]. This region sits on top of two of the transport leads (shown as white triangles). In four point resistance measurements of this sample, we measure the voltage between these two leads and run a current through the other two leads (yellow triangles). (c) A comparison of an ODMR spectrum (green) measured in the laser heated region [green point in (a)] to the spectrum (blue) measured outside the laser heated region [blue point in (a)]. (d) A confocal fluorescence image in the XZ plane of the diamond culet suggests that NV centers created during laser heating extend deeper into the anvil ($\sim 1 \mu\text{m}$) compared to NV centers created by implantation and annealing ($\sim 50 \text{ nm}$). (e,f) A simulation of the expected ODMR spectrum for different thicknesses of the laser-heating-created NV layer. (e) The sample geometry for the simulation. We consider a spherical superconductor of radius $R = 10 \mu\text{m}$ exhibiting the Meissner effect. The superconductor is located a distance $d = 100 \text{ nm}$ from the surface of the diamond culet. A layer of NV centers of uniform concentration is distributed from 10 nm into the culet surface up to a depth t . We assume that all NV centers exhibit a stress induced shift, $\Pi_z = 1.13 \text{ GHz}$ and a stress induced splitting, $\Pi_\perp = 61 \text{ MHz}$ and an applied field of $H_z = 68 \text{ G}$. In addition, we assume a power broadened linewidth of 40 MHz . These parameters correspond to the data point shown in main text Fig. 3(b) (also shown in [f]). We simulate the ODMR spectrum for the thickness, t , valued at 50 nm , $1 \mu\text{m}$, $5 \mu\text{m}$, and $10 \mu\text{m}$. (f) A comparison of the measured data point to the simulated spectra for a layer of NV centers directly below the superconductor. We find good agreement between the data and the simulations for thickness, t , of the NV layer up to $\lesssim 1 \mu\text{m}$. For NV distributions that are $5 \mu\text{m}$ and $10 \mu\text{m}$ deep, we see additional broadening of the ODMR lineshape and a much larger overall splitting. Interestingly, the ODMR lineshape appears bimodal because the stress splitting and the Zeeman splitting add in quadrature. (g-i) A comparison of NV distributions in the implanted NV layer and the laser-heating-created NV layer in sample S1. (h) A XY confocal fluorescence scan of sample S1. We perform XZ confocal scans at two regions denoted by white bars. (g) The first scan, in a region of the sample that was not laser heated, shows the implanted NV layer. Although the implanted NV layer is up to 50 nm deep, the scan exhibits a $10 \mu\text{m}$ region of fluorescence (corresponding to the axial resolution of our microscope). The source of background fluorescence deeper into the diamond anvil is unknown and does not contribute to our measurements. (i) The second scan, in a region of the sample that was laser heated shows a considerably brighter layer of NV centers. Here too, the width of the NV layer appears $\sim 10 \mu\text{m}$ due to the axial resolution of our microscope. Colorbars to the right of each confocal image denotes the fluorescence counts.

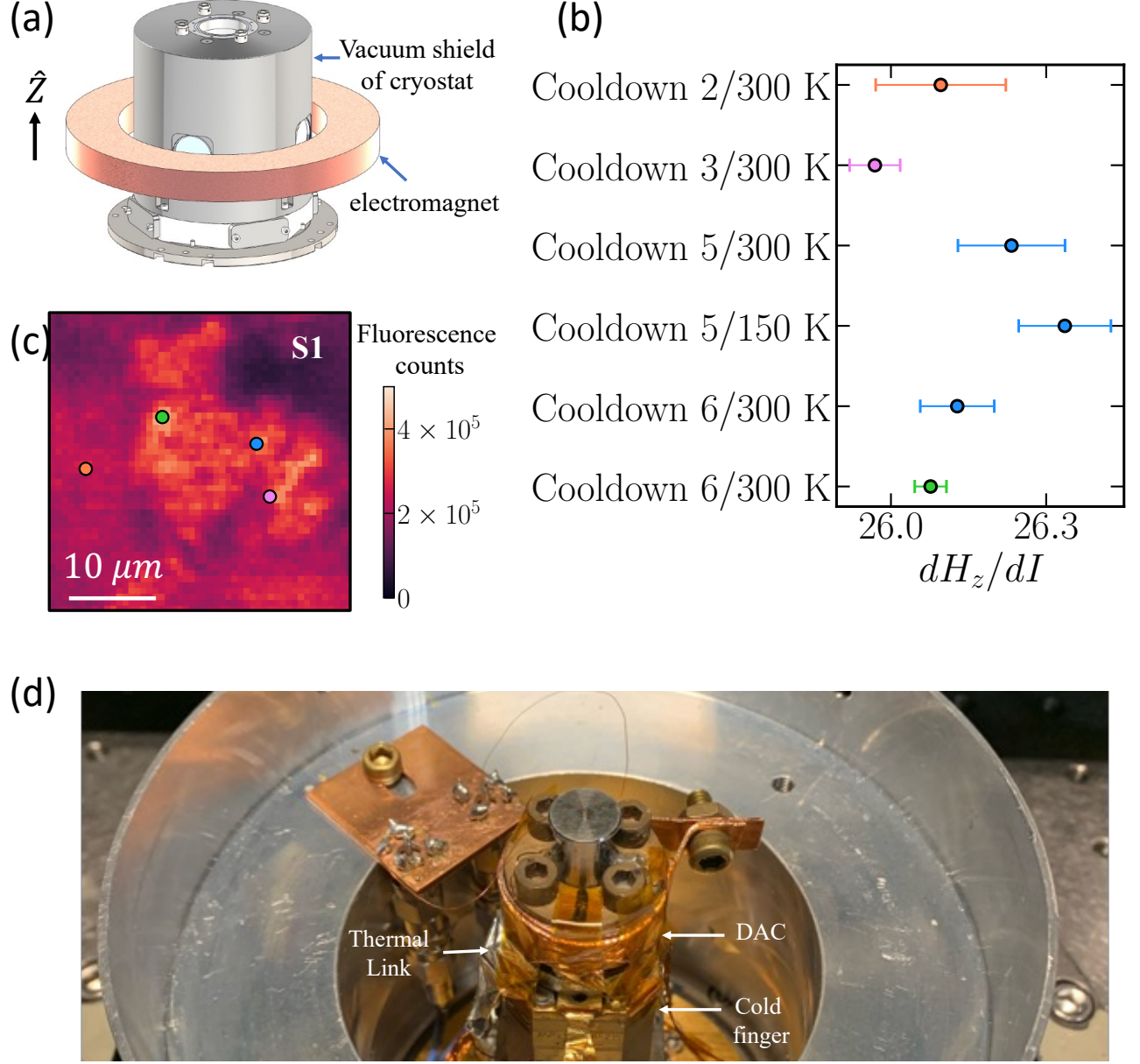


FIG. S4. **Calibration of external magnetic field.** (a) The apparatus used for generating magnetic fields comprises a custom built electromagnet that sits outside the cryostat's vacuum shield. The electromagnet is oriented along loading axis, $\hat{Z}$, which is coincident with the quantization axis of the [111] NV subgroup, $\hat{z}$. Due to minor changes in the placement of the electromagnet during experimental setup, we calibrate the magnetic field at the beginning of every cooldown. (b) A confocal fluorescence image of sample S1; we show local calibrations of H_z at four spatial points depicted by colored circles. **Note:** These spatial points are not connected to those shown in Fig. 4 in the maintext for sample S2. (c) Calibrations at the four spatial locations in sample S1 across four different cooldowns provide similar results for dH_z/dI . The color of the data markers corresponds to the spatial points in (b). For cooldown 5, calibrations in the superconducting region of the sample at two temperatures, $T = 300$ K and $T = 150$ K, yield similar results. (d) Image of the setup in the attodry800 cryostat showing the DAC, cold finger, and thermal link.

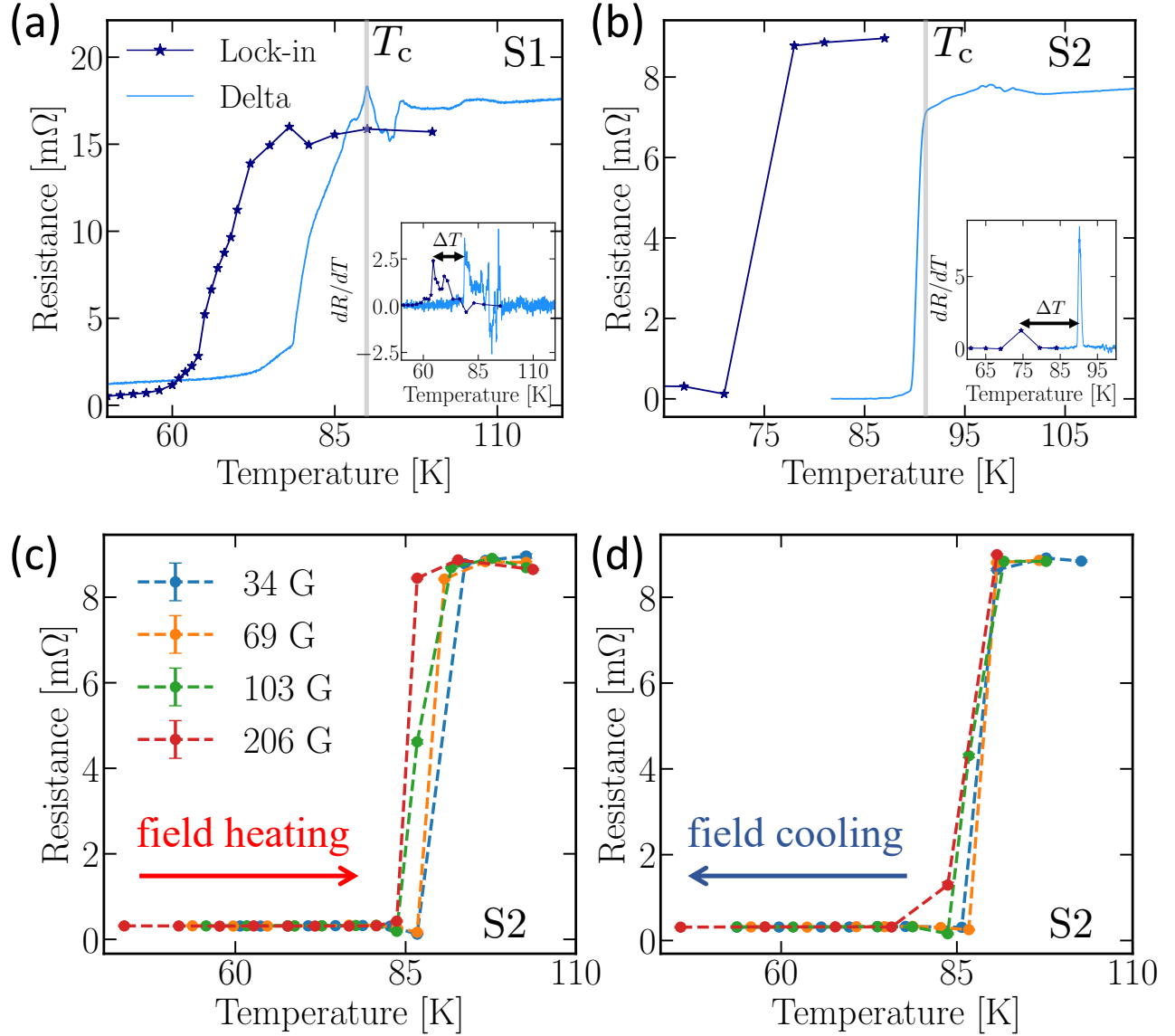


FIG. S5. **Determination of thermal gradients.** Following synthesis, four point resistance of each samples is measured via delta technique [Fig. S2(c,d)] in a cryostat with good sample thermalization [14, 16]. These measurements (shown in light blue data) for samples (a) S1 and (b) S2 are reflective of the true T_c in each case. NV measurements are performed in a dry cryostat where sample thermalization can be challenging; in this cryostat, four point resistance of the sample is measured using lock-in technique (dark blue data). For both samples, the temperature at which we observe a resistive transition in the dry cryostat is lower than the true T_c , suggesting the presence of a temperature gradient (ΔT). We extract ΔT by comparing the derivative (dR/dT) of sample resistance (measured via lock-in and delta techniques) [insets]. Four point resistance of S2 measured via lock-in technique at several values of H_z upon (c) field heating and (d) field cooling. The apparent spread in the transition points observed on field heating suggest that the temperature gradient in the dry cryostat may depend on the direction of the temperature sweep; we attribute this spread to incomplete thermalization of the sample during the sweep. A correction of $\Delta T = 15.7$ K is applied to all data points in (c-d).

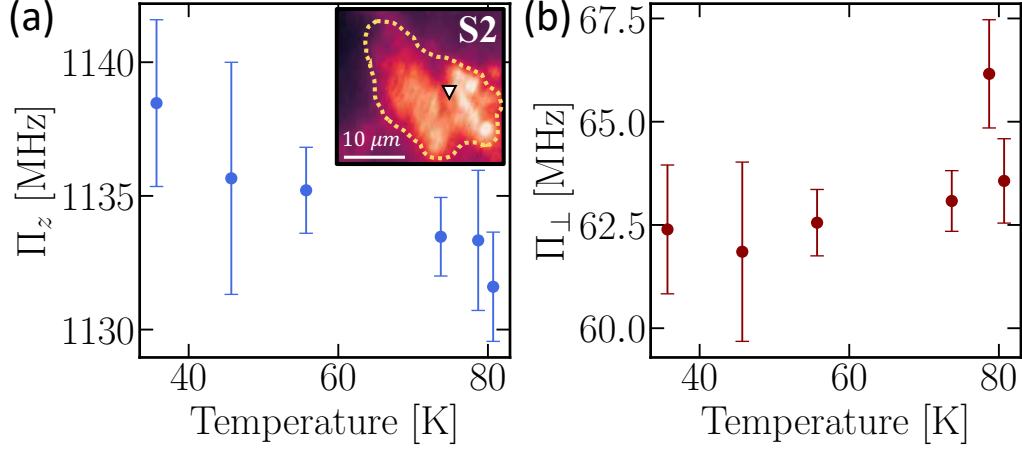


FIG. S6. **Change in stress parameters with temperature in S2.** We directly extract the (a) shift, Π_z , and (b) splitting, $\Pi_\perp$, from ODMR spectra measured at several temperatures at the same spatial location [inverted white triangle marked on the fluorescence scan shown in inset of (a)]. To extract Π_z , we subtract the temperature dependent zero-field splitting, D_{gs} , from the center frequency of the ODMR resonances [19]. We find that Π_z increases with decreasing temperature consistent with an expected increase in sample pressure [13]. For this sample, we observe a comparatively smaller change in the $\Pi_\perp$ stress parameter (within $\sim (2\pi) \times 8$ MHz).